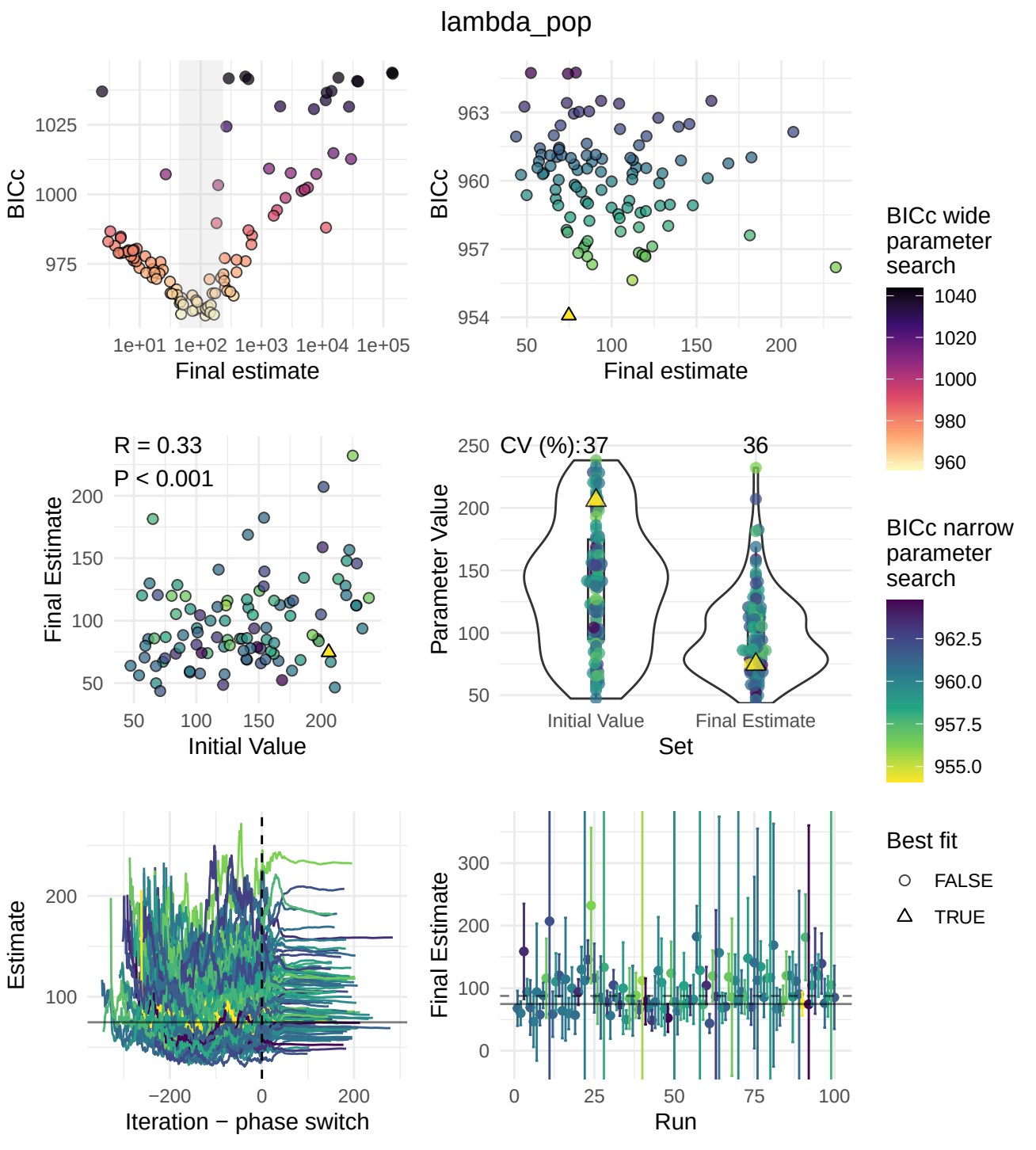
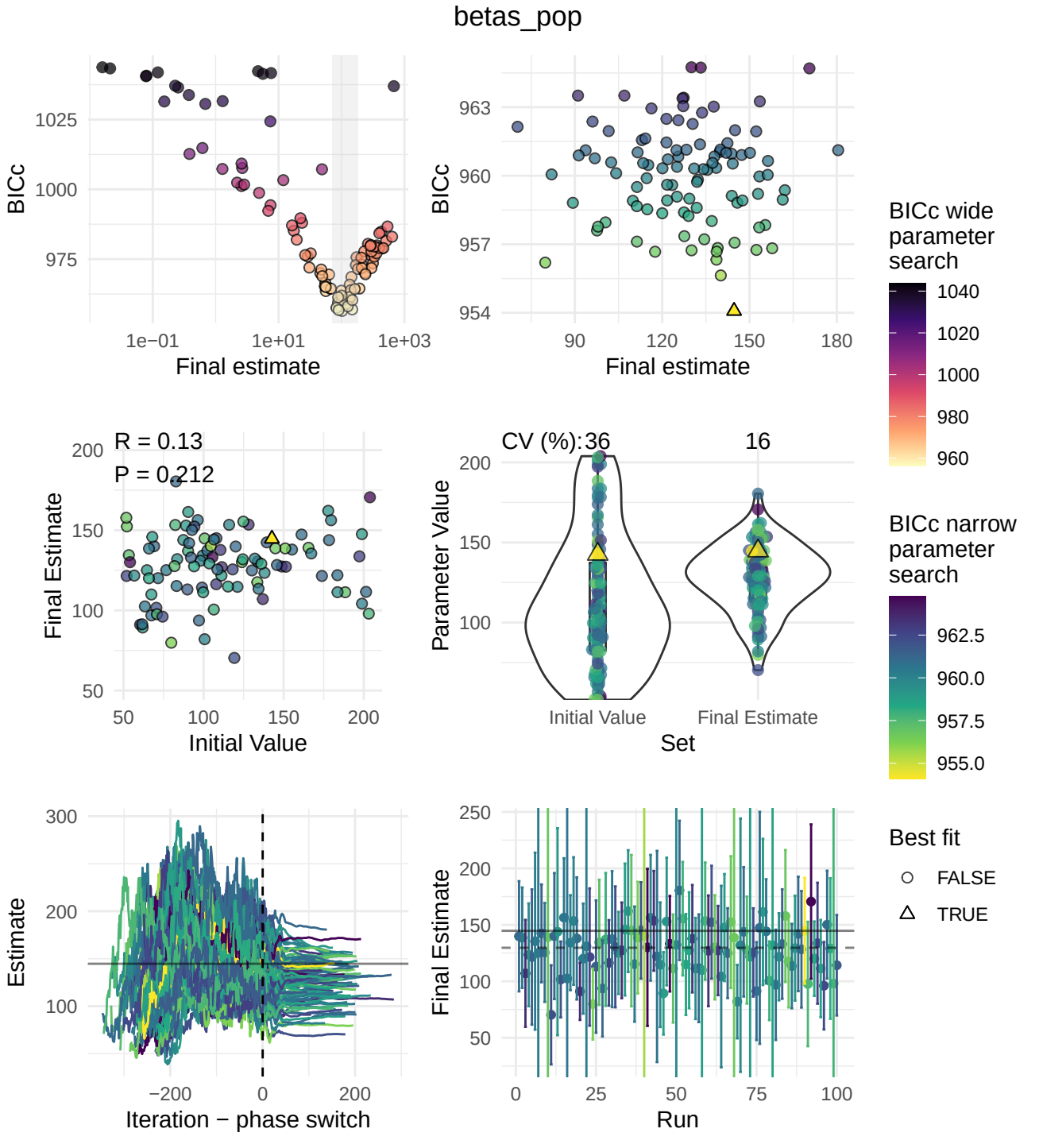
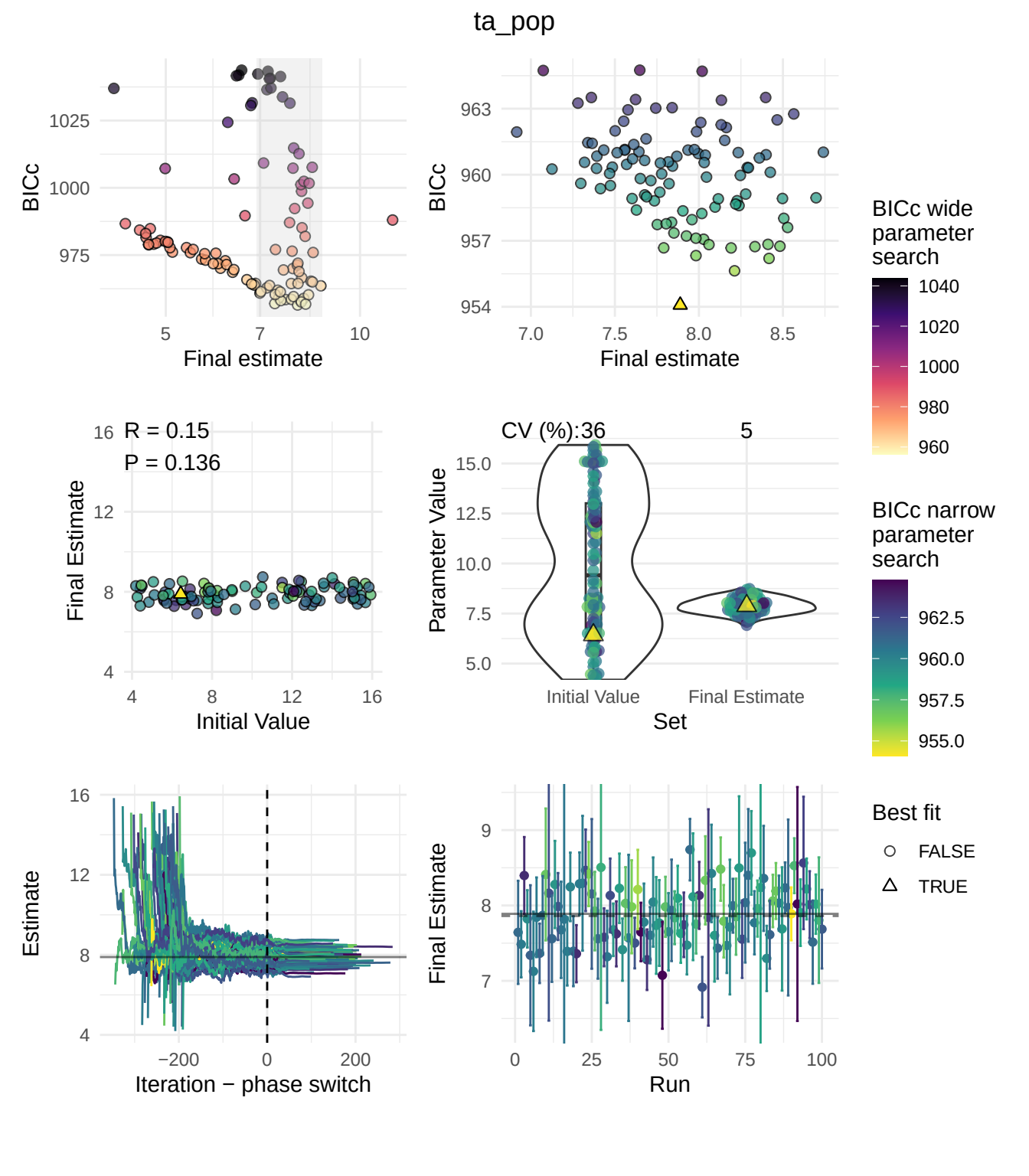
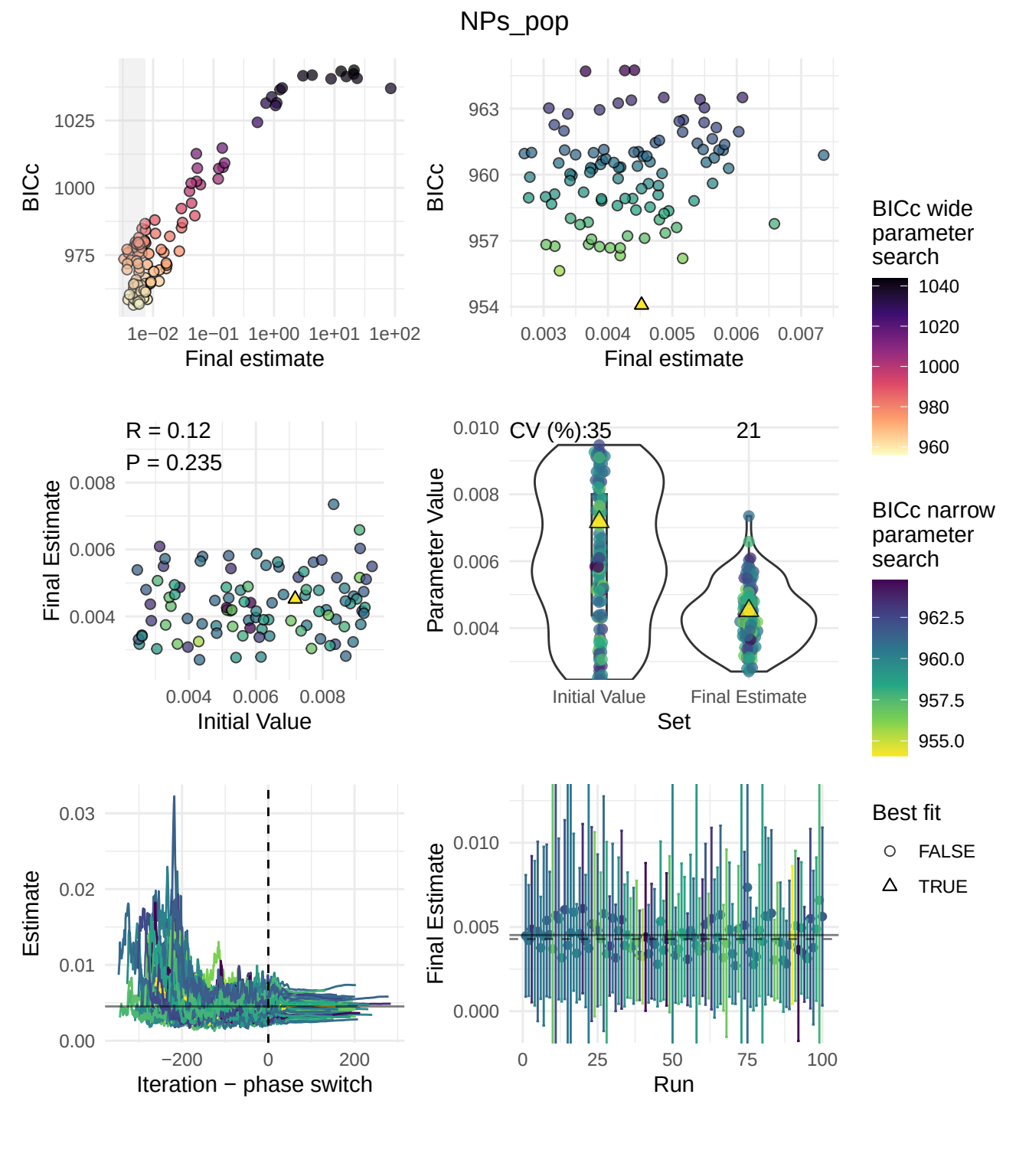
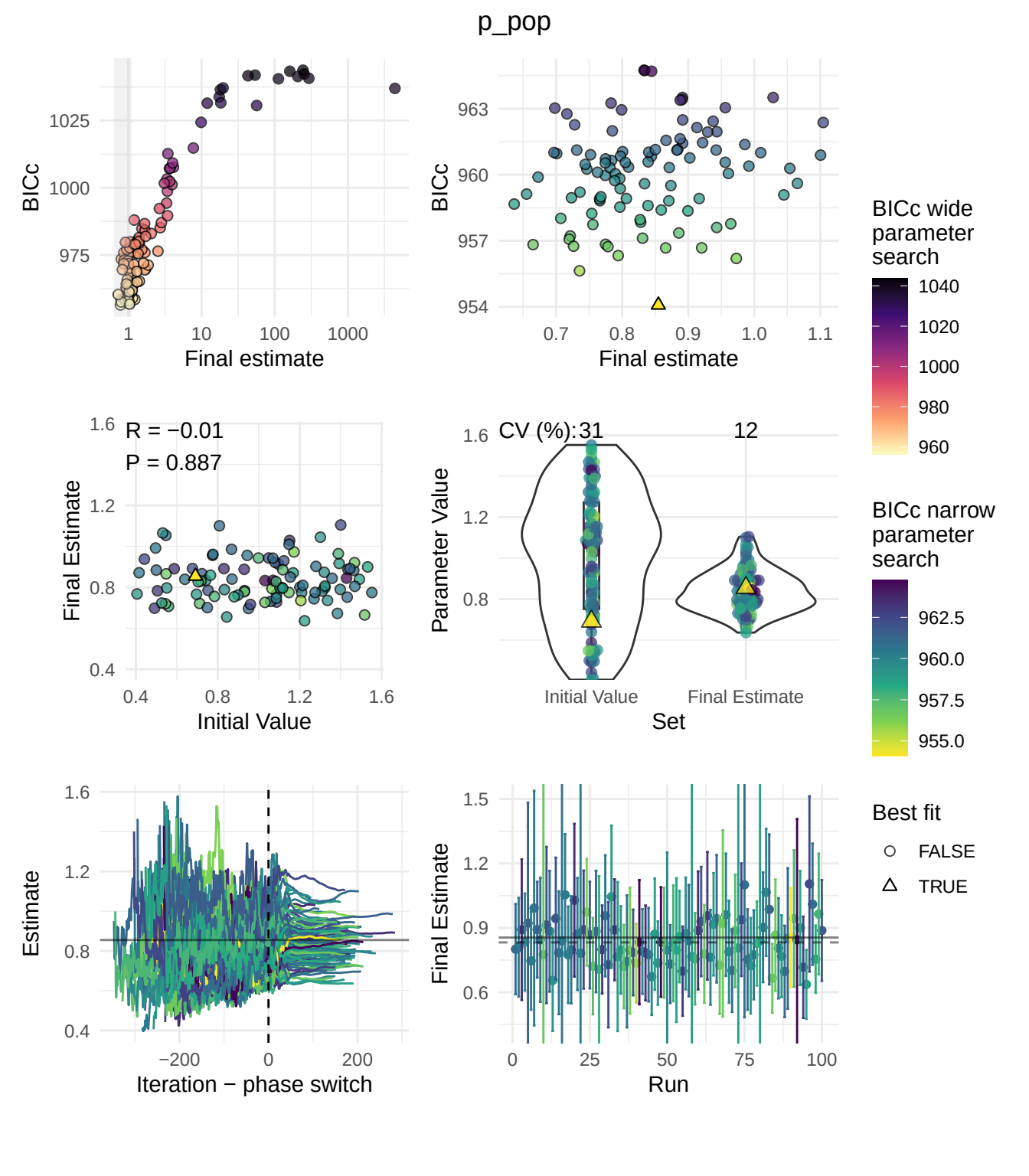
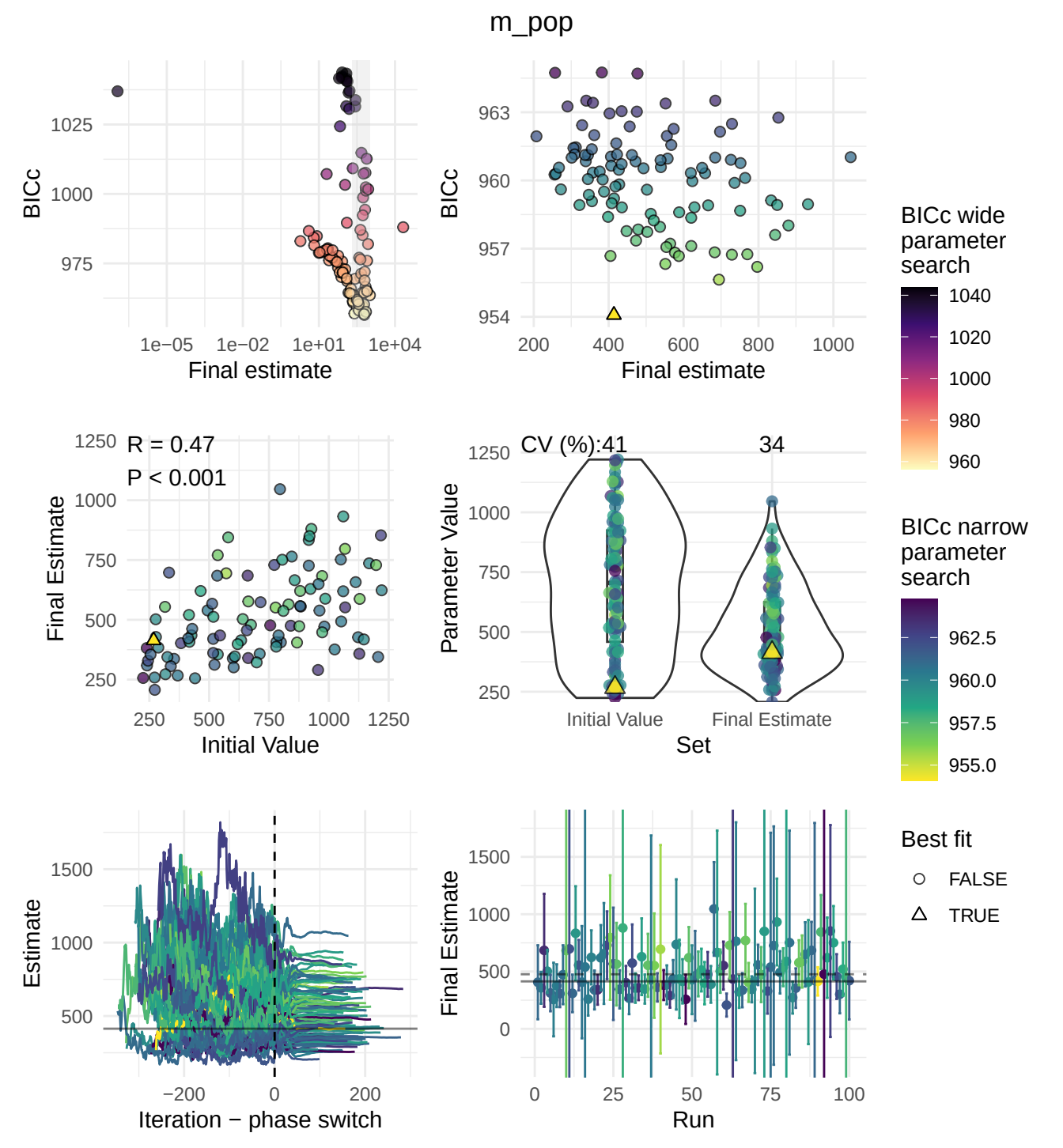

lambda_pop
BICc
1025
1000
975
1e+01 1e+02 1e+03 1e+04 1e+05
Final estimate
BICc
963
960
957
954
50 100 150 200
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
R = 0.33
P < 0.001
Final Estimate
200
150
100
50
50 100 150 200
Initial Value
Parameter Value
CV (%):37
36
250
200
150
100
50
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
200
100
−200 0 200
Iteration − phase switch
Final Estimate
300
200
100
0
0 25 50 75 100
Run
Best fit
FALSE
TRUE

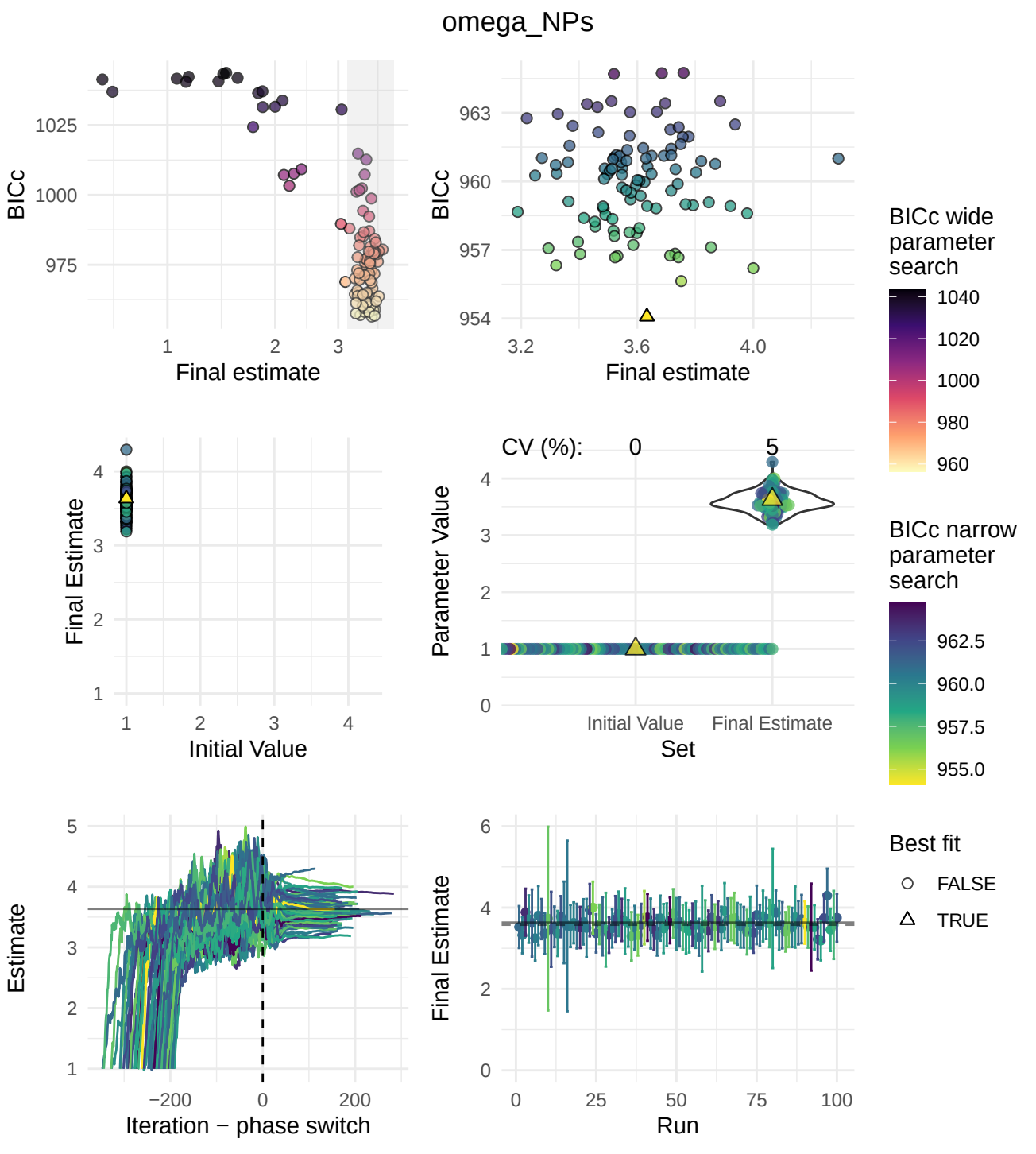

betas_pop
BICc
Final estimate
1025
1000
975
1e−01
1e+01
1e+03
963
960
957
954
90
120
150
180
BICc wide parameter search
1040
1020
1000
980
960
R = 0.13
P = 0.212
Final Estimate
Initial Value
200
150
100
50
CV (%): 36
16
Parameter Value
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
300
200
100
−200
0
200
Final Estimate
Run
250
200
150
100
50
0
25
50
75
100
Best fit
FALSE
TRUE

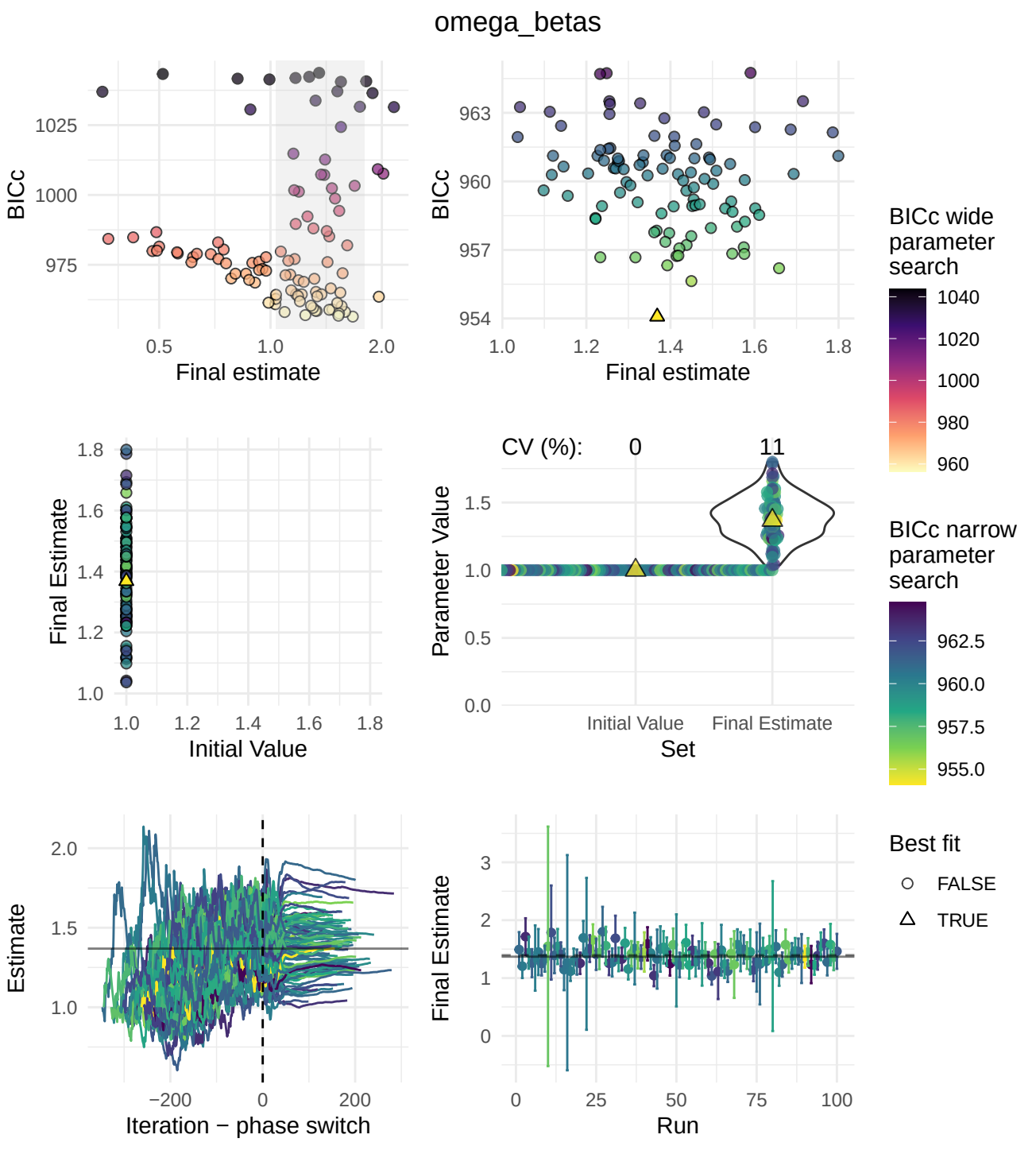

ta_pop
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
BICc narrow parameter search
962.5
960.0
957.5
955.0
R = 0.15
P = 0.136
Final Estimate
Initial Value
CV (%):36
5
Parameter Value
Initial Value
Final Estimate
Set
Best fit
FALSE
TRUE
Estimate
Iteration − phase switch
Final Estimate
Run

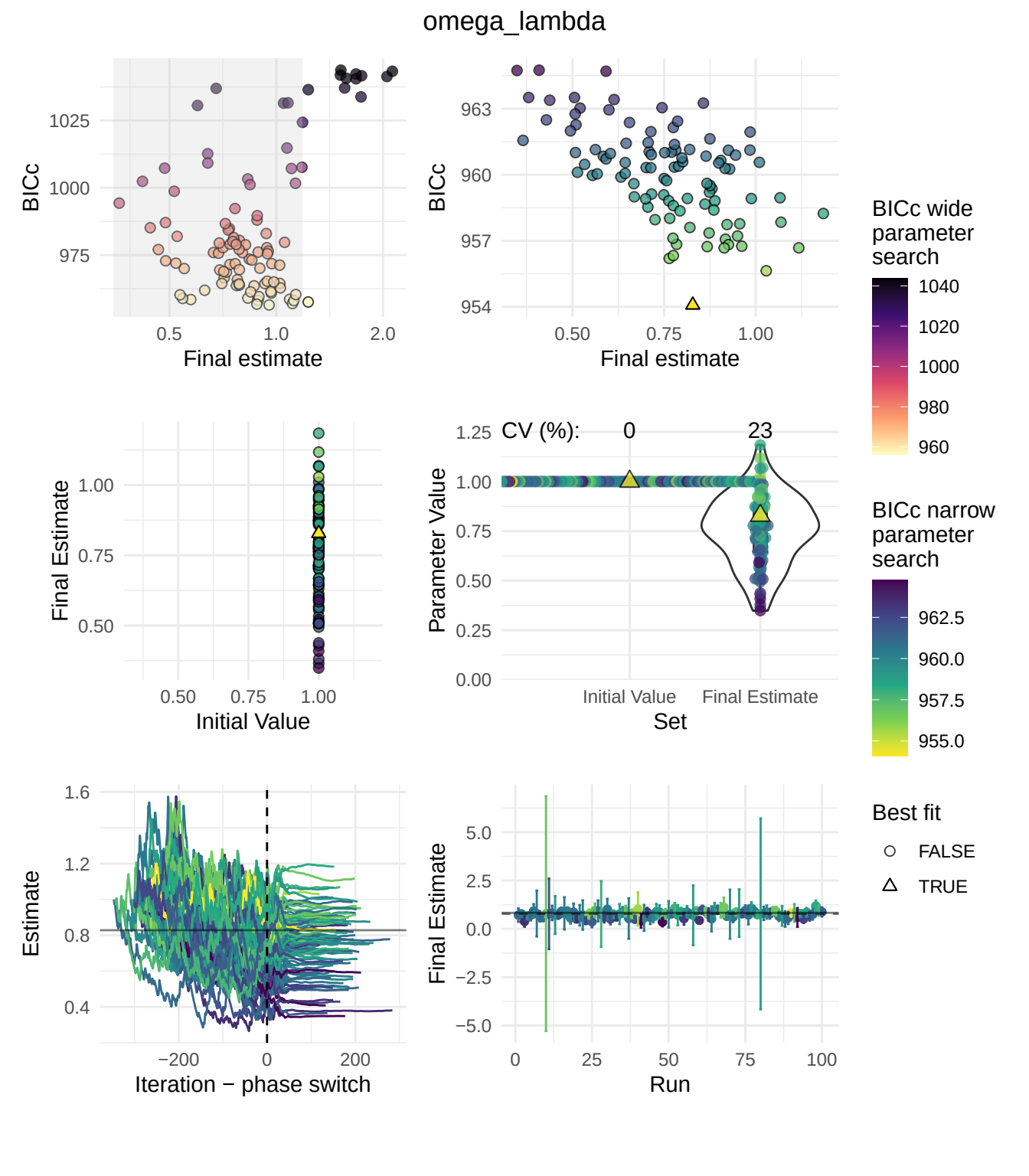

NPs_pop
BICc
Final estimate
1025
1000
975
1e−02
1e−01
1e+00
1e+01
1e+02
963
960
957
954
0.003
0.004
0.005
0.006
0.007
BICc wide parameter search
1040
1020
1000
980
960
R = 0.12
P = 0.235
Final Estimate
Initial Value
0.008
0.006
0.004
CV (%):35
21
Parameter Value
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
0.03
0.02
0.01
0.00
−200
0
200
Final Estimate
Run
0.010
0.005
0.000
0
25
50
75
100
Best fit
FALSE
TRUE

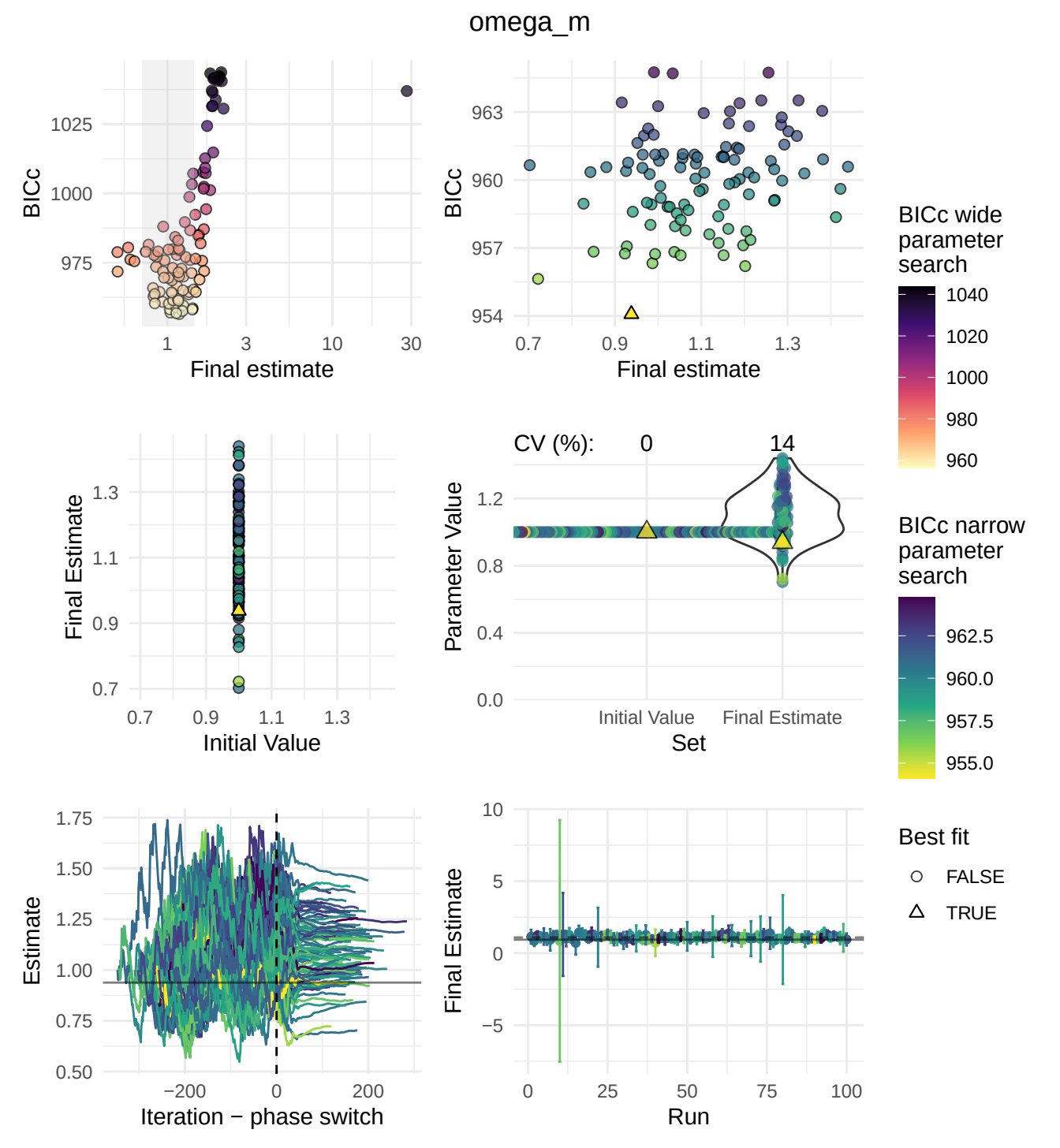

p_pop
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
R = −0.01
P = 0.887
Final Estimate
Initial Value
CV (%):31
12
Parameter Value
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
Final Estimate
Run
Best fit
FALSE
TRUE

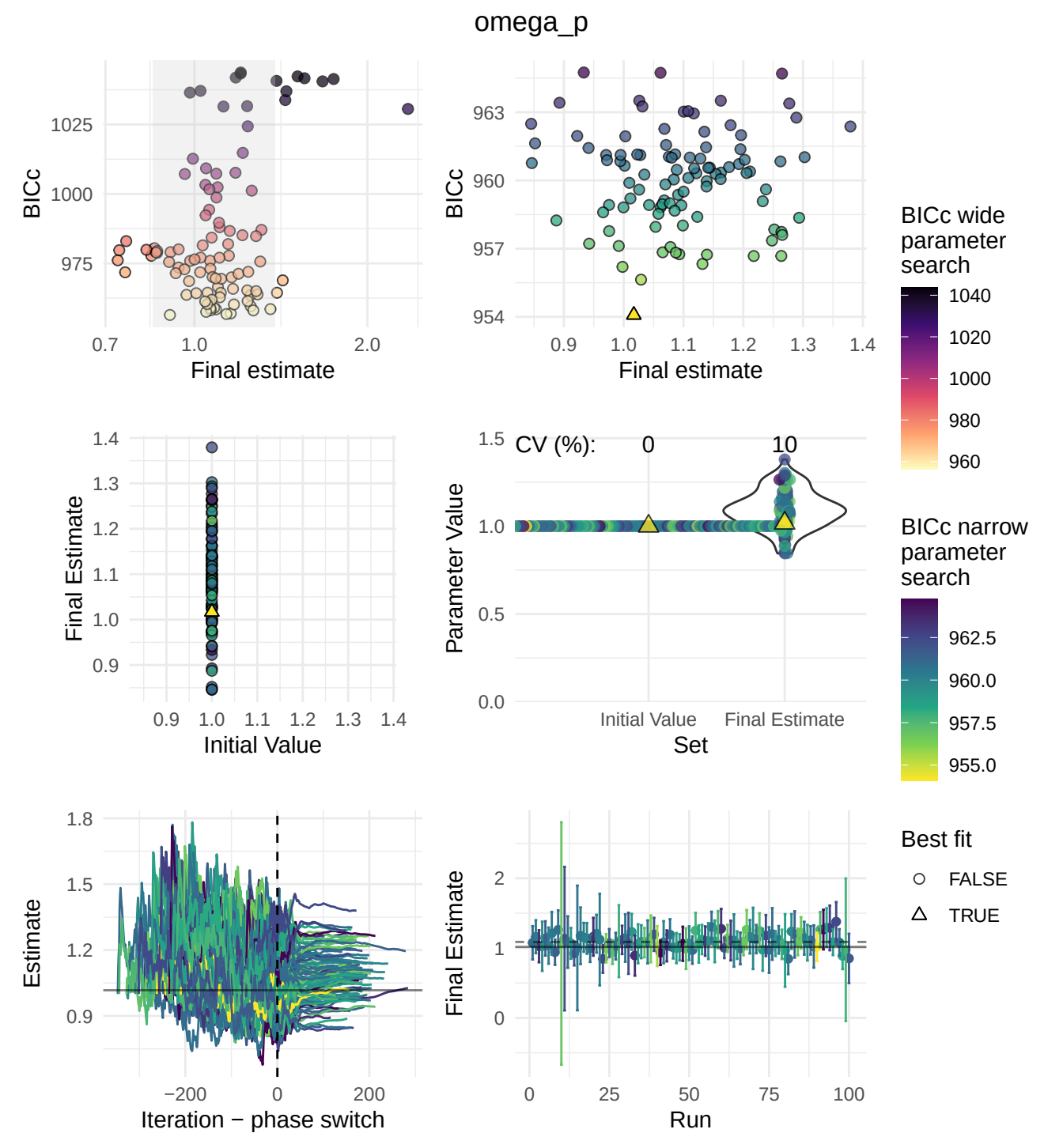

m_pop
BICc
Final estimate
1025
1000
975
1e−05
1e−02
1e+01
1e+04
963
960
957
954
200
400
600
800
1000
BICc wide parameter search
1040
1020
1000
980
960
R = 0.47
P < 0.001
Final Estimate
Initial Value
1250
1000
750
500
250
CV (%):41
34
Parameter Value
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
1500
1000
500
−200
0
200
Final Estimate
Run
0
25
50
75
100
Best fit
FALSE
TRUE

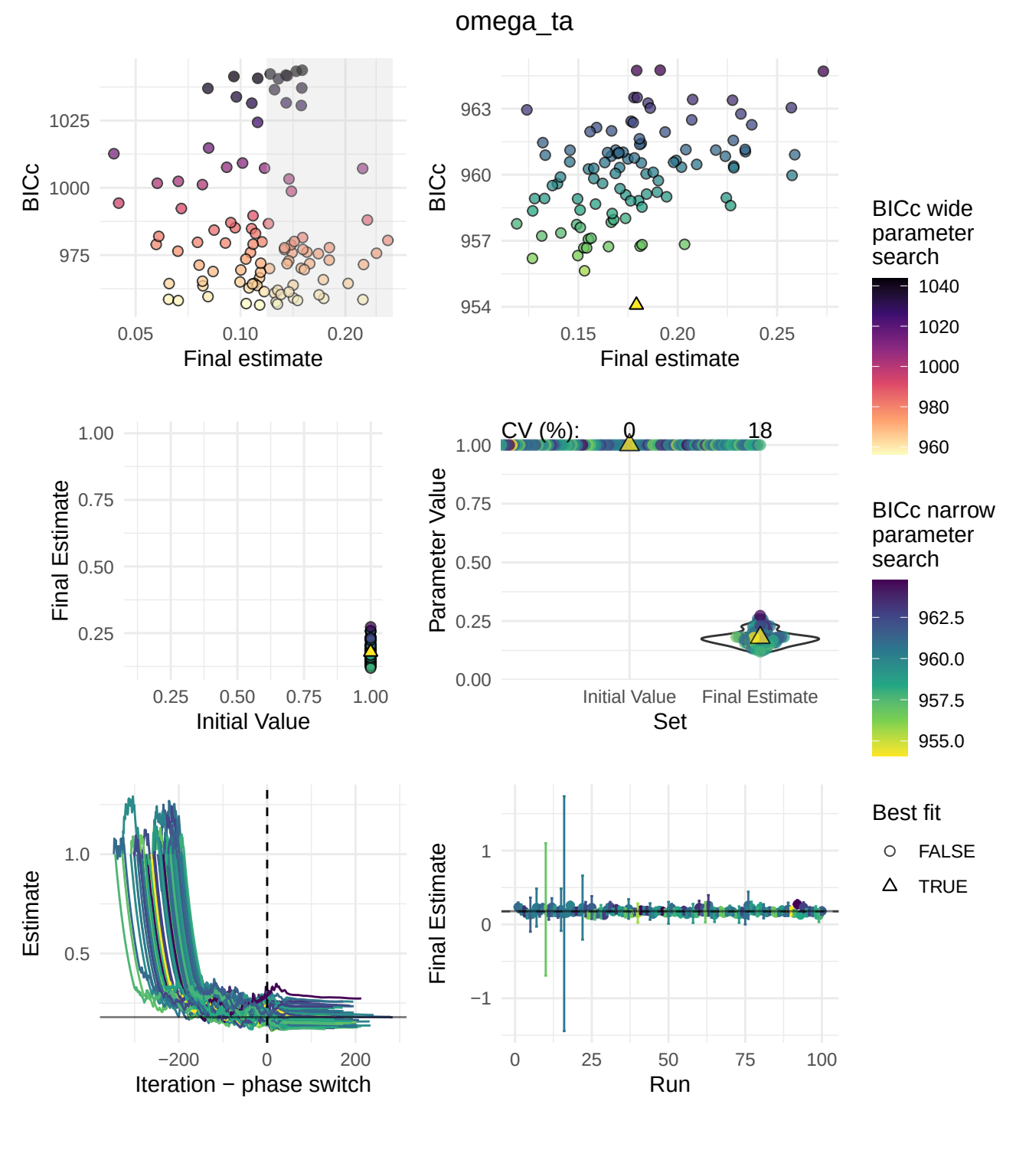

omega_NPs
BICc
1025
1000
975
1
2
3
Final estimate
BICc
963
960
957
954
3.2
3.6
4.0
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
Final Estimate
4
3
2
1
1
2
3
4
Initial Value
CV (%):
0
5
Parameter Value
4
3
2
1
0
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
5
4
3
2
1
−200
0
200
Iteration − phase switch
Final Estimate
6
4
2
0
0
25
50
75
100
Run
Best fit
FALSE
TRUE

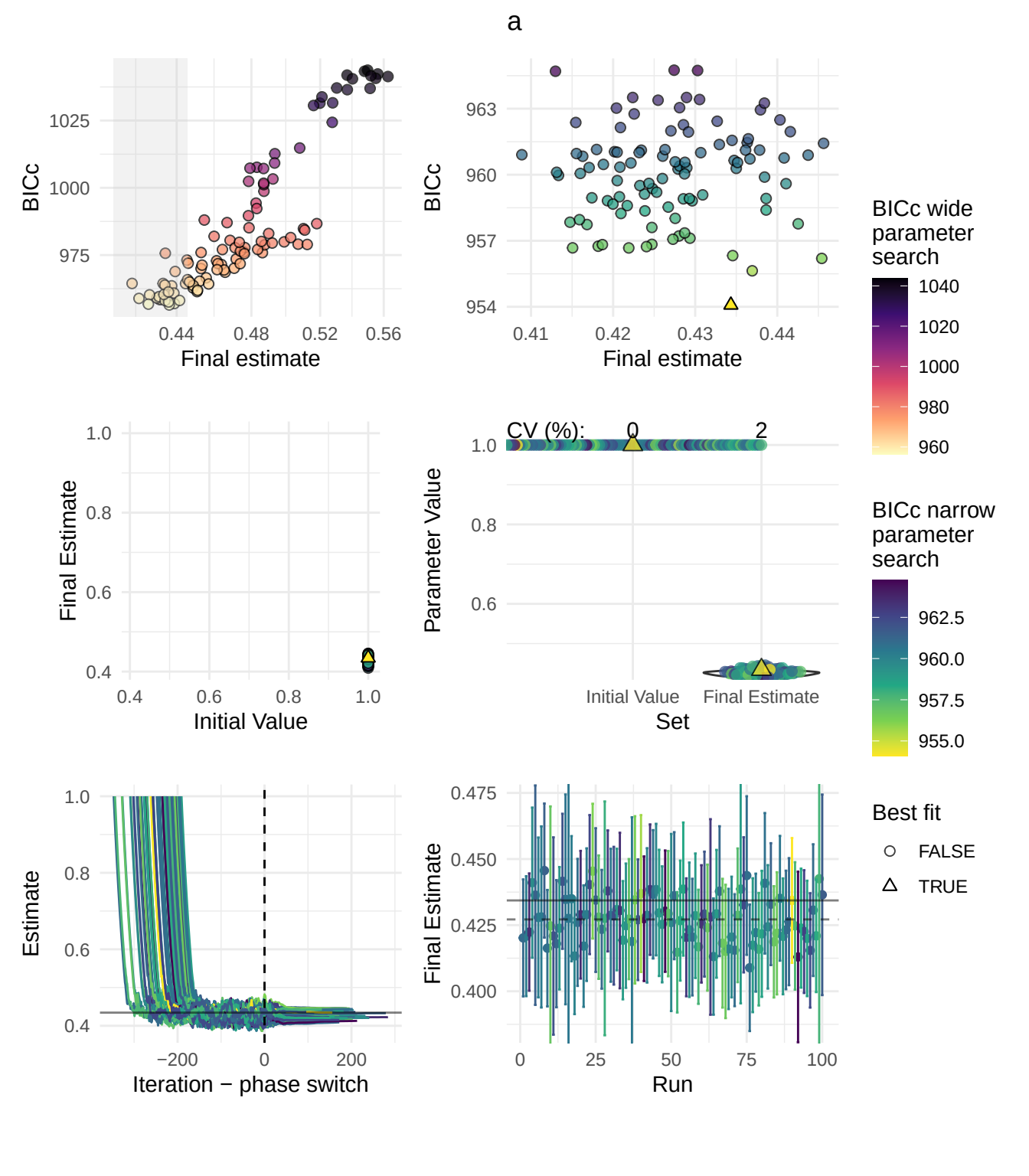

omega_betas
BICc
Final estimate
BICc wide parameter search
BICc narrow parameter search
Final Estimate
Initial Value
CV (%): 0 11
Parameter Value
Initial Value
Final Estimate
Set
Estimate
Iteration − phase switch
Run
Best fit
FALSE
TRUE


omega_lambda
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
Final Estimate
Initial Value
CV (%):
0
23
Parameter Value
Set
Initial Value
Final Estimate
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
Best fit
FALSE
TRUE
Run


omega_m
BICc
1025
1000
975
1
3
10
30
Final estimate
BICc
963
960
957
954
0.7
0.9
1.1
1.3
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
Final Estimate
1.3
1.1
0.9
0.7
0.7
0.9
1.1
1.3
Initial Value
CV (%):
0
14
Parameter Value
1.2
0.8
0.4
0.0
Initial Value
Final Estimate
Set
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
1.75
1.50
1.25
1.00
0.75
0.50
−200
0
200
Iteration − phase switch
Final Estimate
10
5
0
−5
0
25
50
75
100
Run
Best fit
FALSE
TRUE


omega_p
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
BICc narrow parameter search
962.5
960.0
957.5
955.0
Final Estimate
Initial Value
CV (%):
0
10
Parameter Value
Set
Initial Value
Final Estimate
Best fit
FALSE
TRUE
Estimate
Iteration − phase switch
Run


omega_ta
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
BICc narrow parameter search
962.5
960.0
957.5
955.0
Final Estimate
Initial Value
CV (%):
0
18
Parameter Value
Set
Initial Value
Final Estimate
Best fit
FALSE
TRUE
Estimate
Iteration − phase switch
Run


a
BICc
Final estimate
BICc wide parameter search
1040
1020
1000
980
960
Final Estimate
Initial Value
CV (%):
0
2
Parameter Value
Set
Initial Value
Final Estimate
BICc narrow parameter search
962.5
960.0
957.5
955.0
Estimate
Iteration − phase switch
Final Estimate
Run
Best fit
FALSE
TRUE